

Blood biomarker discovery summary

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This synthetic-data analysis assessed a blood-based panel of up to five methylation and three protein markers. The full methods and results are in *Biomarker-discovery-and-modeling.html*.

QC findings

- repeated draw: The dataset contains 350 samples from 300 patients. Each patient’s earliest draw was kept so patients with two draws would not receive twice the weight of patients with one draw during modeling.
- missing protein: Site C contributed 95 samples from 80 patients but had no protein data. I excluded these 80 patients from protein-based model training because the required predictors were unavailable.
- missing data: Three protein features were missing in more than 20% of assayed patients, and one sample was missing more than 20% of protein features. I removed them to focus on reliably measured markers and samples. No methylation region exceeded 5% missingness, and the fragmentomics features used here were complete.
- impute missing data: Missing methylation measurements reflect insufficient coverage rather than a biological value of zero. I used the median of each feature’s training values because it is less sensitive to extreme values. Missing protein measurements were below the lower limit of quantification (LLOQ). Because the LLOQs were unavailable, I used half the smallest positive training value for each protein as a low-concentration proxy. Imputation was refitted within each training fold to avoid data leakage.
- no outlier: I inspected PCA plots for all three modalities to identify unusual samples that might distort the analysis; there were no obvious outliers in the first two components.
- protein value: $\log_2(x)$ transformation after positive-value imputation during modeling to reduce concentration skew. The modeling features were then standardized so the LASSO penalty was comparable across markers.
- batch effects: PCA suggested site differences in methylation and plate differences in protein measurements. I used control-only linear models, adjusting for age and sex, to identify and subtract significant site and plate effects. Plate separation largely disappeared after correction, but some methylation separation by site remained and warrants investigation.

Modeling

- Marker selection: The modeling cohort has 219 patients and more than 2,000 molecular features, while the intended panel allows at most five methylation and three protein markers. I used LASSO to select markers while fitting the model, favoring predictors that add information alongside those already selected rather than screening each marker separately for association with cancer before modeling.
- Nested cross-validation: Inner folds selected the LASSO penalty and fitted preprocessing within their training data. The untouched outer folds then evaluated the complete selection and prediction procedure, limiting data leakage and giving a held-out performance estimate.
- Repeated modeling: I repeated the nested cross-validation five times with different fold splits to assess how sensitive performance and marker selection were to the split. Selection frequency was calculated across the outer training folds; the repeats reuse the same patients and are not independent validation cohorts.

Results

Original methylation + protein model

The table below shows mean held-out performance across five outer-CV repeats. Sensitivities use empirical 95%-specificity cutoffs estimated from held-out controls within each site group. Overall performance is broadly similar at

sites A and B. Pooled sensitivity rises with cancer stage, and stage I sensitivity is only 0.200, suggesting that the model detects later-stage cancers more readily.

Table 1. Outer-CV performance. Ctrl and Case are mean counts; SD is the AUC standard deviation. All and I–IV are sensitivities at 95% specificity.

Site	Ctrl	Case	AUC	SD	All	I	II	III	IV
All	106	113	0.890	0.018	0.573	0.200	0.347	0.572	0.800
A	31	78	0.884	0.016	0.562	0.114	0.314	0.593	0.764
B	75	35	0.892	0.030	0.491	0.280	0.360	0.429	0.727

The final full-cohort panel (Table 2) contains the following markers. Beta means are average methylation for cancer and control patients in the entire modeling cohort; dashes indicate fields that do not apply to proteins. High frequencies suggest that these markers were selected consistently under this resampling scheme. All five selected methylation regions either have high GC content or overlap repeats, complicating ddPCR assay design.

Table 2. Final panel: raw methylation means, selection frequency across 25 outer training folds, and region annotations.

Marker	Freq	Beta_Cancer	Beta_Control	Locus	Gene	GC	Repeat
CpG_R0607	100%	0.093	0.050	chr22:37224754-37225135	GENE0044	0.589	Yes
CpG_R0164	96%	0.121	0.080	chr3:14610606-14610834	GENE0980	0.867	No
CpG_R0805	88%	0.176	0.136	chr14:92762010-92762347	GENE3061	0.854	No
CpG_R1263	84%	0.055	0.023	chr2:93875211-93875432	GENE3373	0.835	No
CpG_R0360	80%	0.389	0.340	chr10:4478064-4478328	GENE1161	0.870	No
CEA	100%	-	-	-	-	-	-
CYFRA21_1	84%	-	-	-	-	-	-
GDF15	68%	-	-	-	-	-	-

methylation + protein model after removing unfeasible methylation regions

Removing regions that overlap repeats or have GC content outside [0.4, 0.6] retained 848 of 2,000 methylation regions. Using the same outer folds, pooled AUC fell from 0.890 to 0.761; sensitivity at 95% specificity fell from 0.573 to 0.271. The filter removed several strong predictors in this cohort.

Adding fragmentomics

Because fragmentomics features have been associated with cancer in the literature, I tested whether they could improve performance by adding the short-to-long fragment ratio, mitochondrial fraction, mean length, nucleosome periodicity, and 4-mer end motifs to the original inputs. Pooled AUC did not improve: 0.888 versus 0.890. The final fit retained the short-to-long fragment ratio from this modality with high selection frequency, consistent with its reported association with cancer.

Next steps

Given the above results and observations, here are some actions next:

- Find cohorts with more stage I/II samples: since the product is eventually for detecting early cancers, using early stage samples can help find such markers, rather than the markers associated with late stages
- Check methylation pattern at the fragment level, current methylation is the average beta for a region, not informative on which CpGs in the region are(un)methylated for cancers, preventing designing ddPCR probes
- Check methylation regions with genetic polymorphisms, so to avoid those regions for better ddPCR assay design
- Review results with lab scientists to see whether ddPCR assays can be designed from identified regions.
- Compare early- and late-stage cancers to identify candidate markers of progression, then evaluate those markers in longitudinal samples before using them to monitor disease over time.